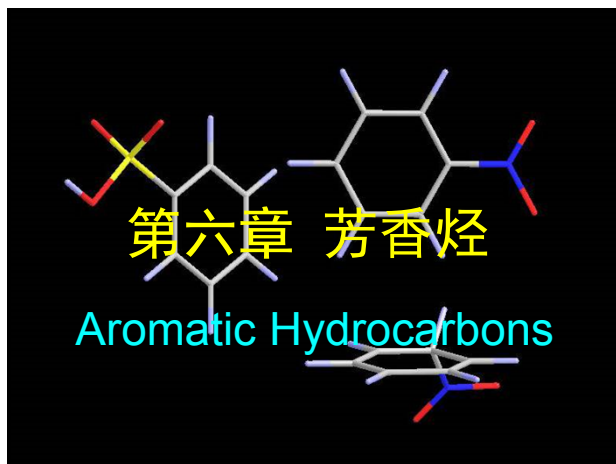




CONTENT	
1	芳香烃的分类、结构及命名
2	芳香烃的物理性质(自学)
3	芳香烃的化学性质
4	苯环上亲电取代的定位规律
5	稠环芳烃

2

何为芳香烃？

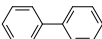
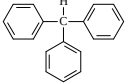
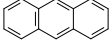
- 含有苯环结构的烃称为芳香烃,简称芳烃 **Aromatic hydrocarbon**。芳香烃的名称来源于早期发现来自香树脂、香精油等具有芳香气味的天然产物。
- 具有芳香族化合物性质,但没有苯环结构的化合物称为非苯芳烃 (nonbenzenoid aromatic hydrocarbon)

3

6.1 芳香烃的分类、结构及命名

6.1.1 芳香烃的分类

芳香烃

- 单环芳烃:  
- 多环芳烃:  
- 稠环芳烃:  
- 非苯芳烃: 

不含苯环,但具有苯的特性“芳香性”的碳环化合物

4

苯的结构

苯的特征:

- 分子式 C_6H_6 , C:H=1:1, 不饱和度为4
- 具有特殊稳定性和对化学反应的惰性

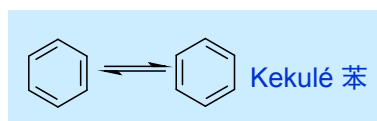
芳香族化合物的特性 —— 芳香性

- 苯环易发生取代, 难发生加成反应
- 不易氧化
- 苯环具有特殊稳定性

5

最为人们所普遍接受的结构

共轭环己三烯苯——Kekulé 苯

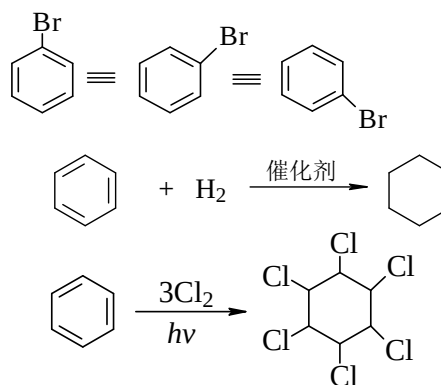


6

Kekulé 苯能解释苯的一些现象:

- (a) 苯的一取代物只有一种
- (b) 苯可以加氢还原为环己烷
- (c) 在光照条件下, 苯可以和3分子氯气加成

7



8

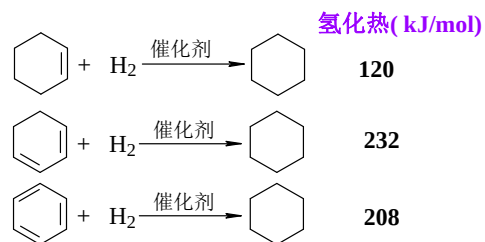
Kekulé苯结构式不能解释的现象：

- 苯容易发生取代反应，却难于发生加成和氧化反应
- 按照 **Kekulé** 结构式，邻位二取代苯应该有两个异构体，但实际上只有一个
- 苯的氢化热远低于环己三烯结构



9

环烯烃的氢化热



$$120 \times 3 = 360 \text{ (kJ/mol)}$$

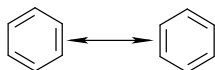
$$360 - 208 = 152 \text{ (kJ/mol)} \text{ —— 苯的稳定化能}$$

10

最为人们所普遍接受的结构

共轭环己三烯苯——Kekulé 苯

- 近代物理方法对苯结构的研究结果
 - 苯的6个碳原子形成平面正六边形
 - 碳-碳键长均等，为 140 pm
 - 所有键角均为120°
- 根据共振论，苯可视为2个等价的Kekulé结构式的共振杂化体：

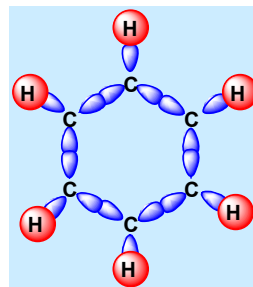


Robinson 1925年
建议用如下表达式



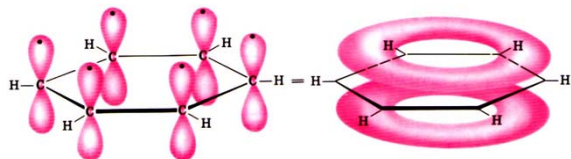
苯的分子轨道描述

- 6个碳原子都是sp²杂化



苯的分子轨道描述

- 未参与杂化的 p 轨道带着一个电子垂直于环平面，形成一个闭合环状的大 π 键



- 苯的6个碳原子形成平面正六边形
- 碳-碳键长均等，为 1.40 Å
- 所有键角均为 120°

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6.1.3 苯及其衍生物的命名和异构

1. 单取代苯的命名

苯环作为母体：—X，—NO₂，—NO，简单烷基



氯苯



硝基苯



亚硝基苯



甲苯

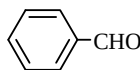
Chlorobenzene nitrobenzene nitrosobenzene methylbenzene (Toluene)

苯环作为取代基：R之前基团与苯环相连时

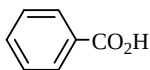
—COOH —SO₃H —COOR —COX —CONH₂

—CN —CHO >C=O —OH —NH₂ —OR —R

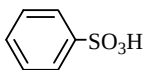
—OH, —NH₂与苯环一起构成母体“苯酚”、“苯胺”



苯甲醛



苯甲酸

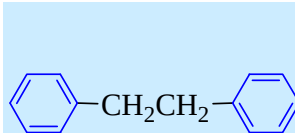


苯磺酸

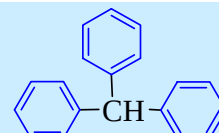
15

苯环作为取代基

分子中有多个苯环，
以任一苯环为母体命名都有困难时



1,2-二苯基乙烷



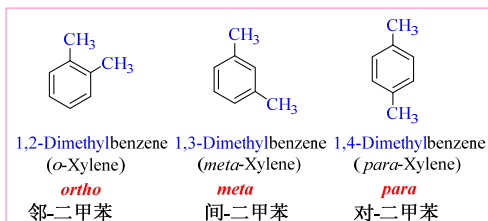
三苯基甲烷

16

二取代苯的命名

- 2个取代基相同

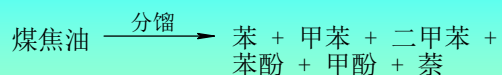
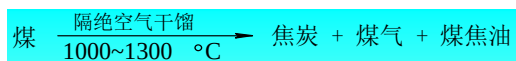
(1,2- *o*- 邻; 1,3- *m*- 间; 1,4- *p*- 对)



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芳香烃的来源(了解)

1. 煤焦油的分馏



煤焦油大约有350种不同的化合物。从1000吨的煤炼焦中，可得6.5吨苯、0.6吨甲苯和二甲苯

18

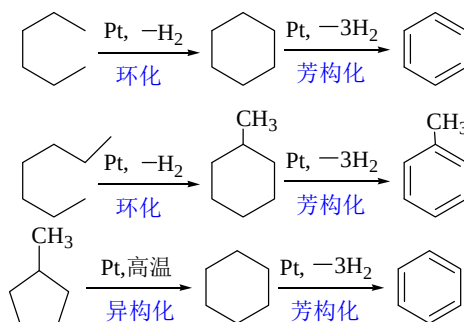
- 在分馏中有约40%的焦油经分馏蒸出

- (1) 80~170℃ 轻油(1~2%) 苯、甲苯、二甲苯
- (2) 170~215℃ 酚油(2~3%) 苯酚、甲酚、二甲酚
- (3) 215~230℃ 萘油(8~10%) 萘
- (4) 230~290℃ 洗涤油(8~10%) 萘、蒽、芴
- (5) 290~350℃ 蒽油(18~25%) 蒽、菲

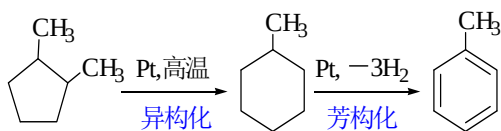
- 残留物 焦油沥青(约55%)

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2. 石油的芳构化

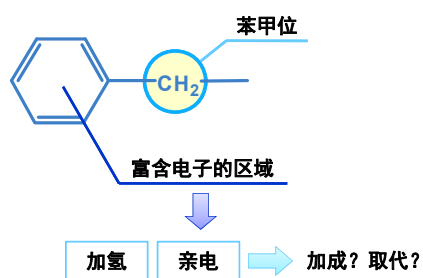


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芳香烃的化学性质



芳香烃的化学性质

芳香亲电取代反应

加成反应

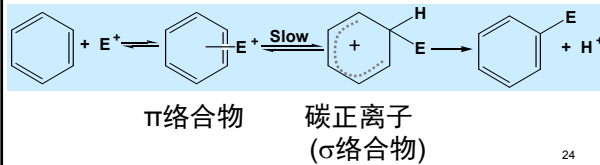
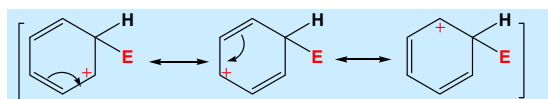
氧化反应

α -卤代反应

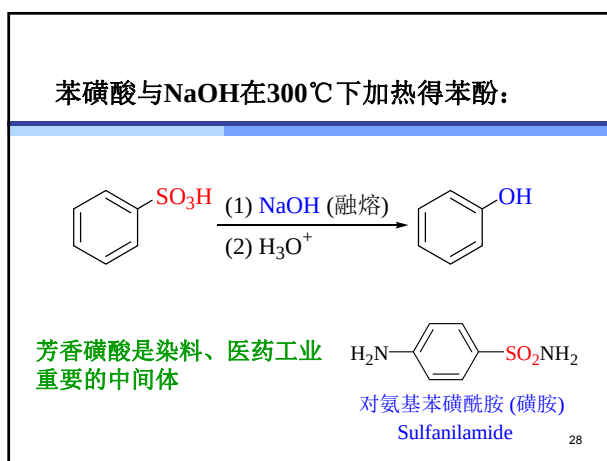
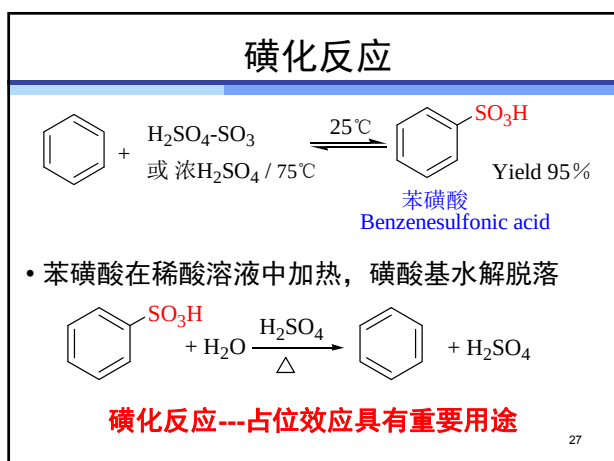
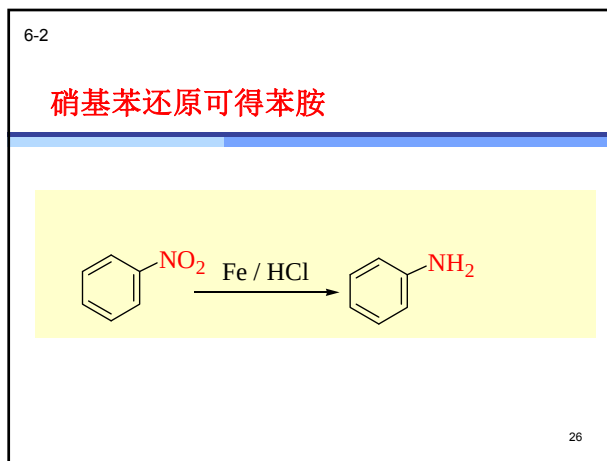
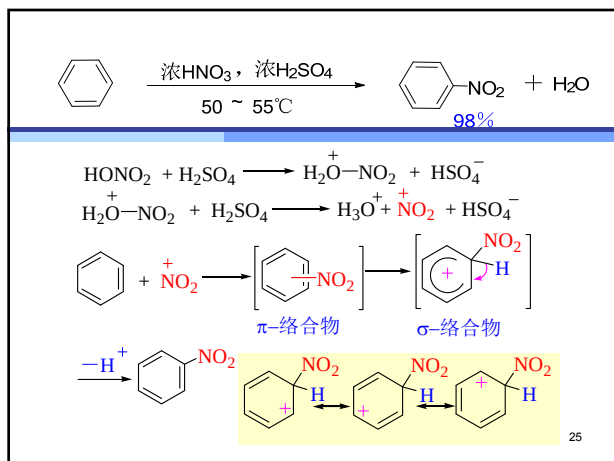
23

芳香亲电取代反应

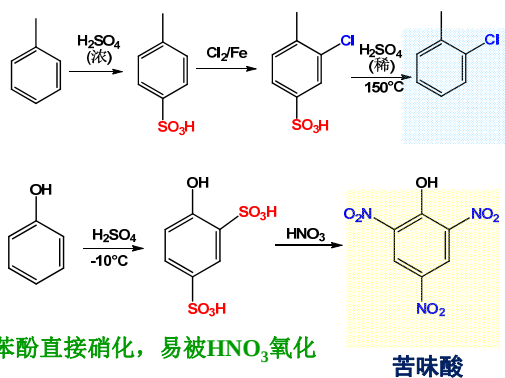
(Electrophilic Aromatic Substitution)



24

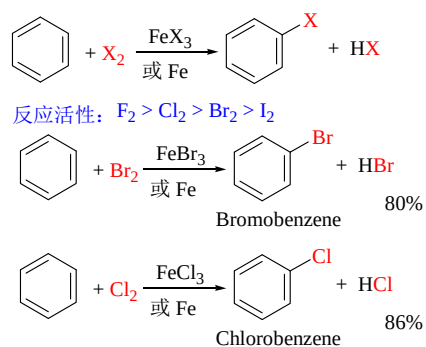


利用磺化反应的可逆性，保护芳环某些特定位置



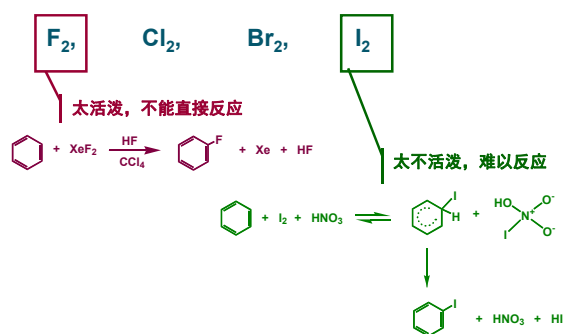
29

卤化反应

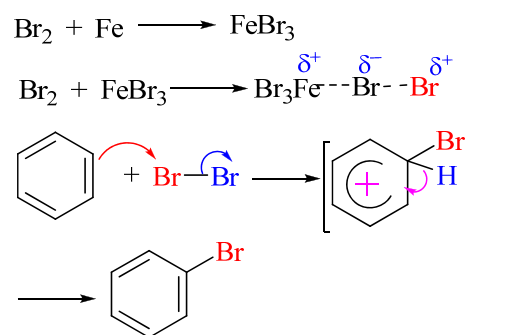


30

反应活性:



反应机理



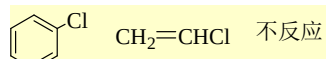
32

傅—克（Friedel—Crafts）反应

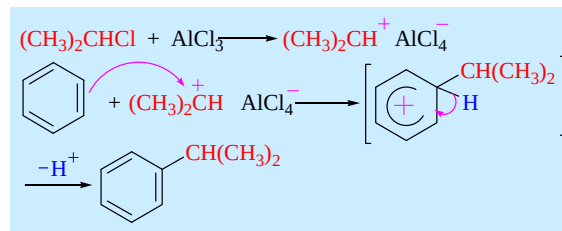
(A) 烷基化(Alkylation)



催化剂 Lewis 酸
 FeCl_3 , SnCl_4 , ZnCl_2
 HF , H_2SO_4 , H_3PO_4 , BF_3

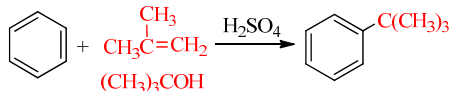
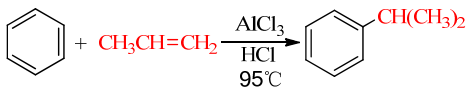
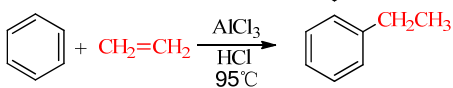
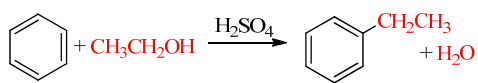


反应机理



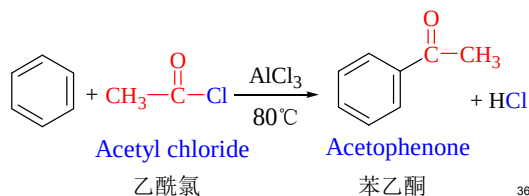
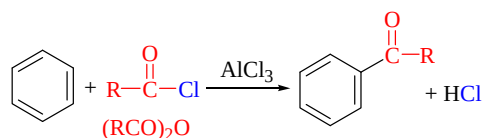
34

烷基化试剂除卤代烃外，还可以用醇、烯烃



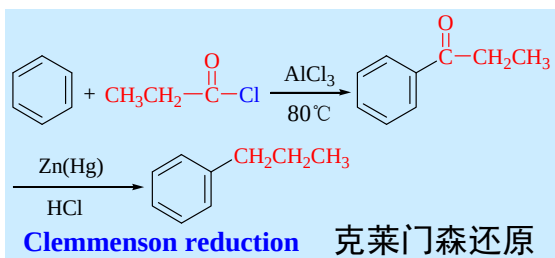
35

(B) 酰基化(Acylation)

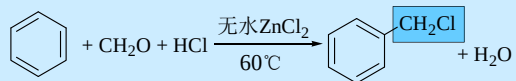


36

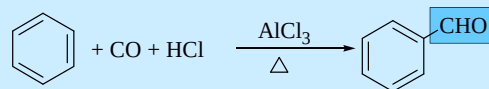
酰基化—还原反应系列



37



氯甲基化 (Chloromethylation)

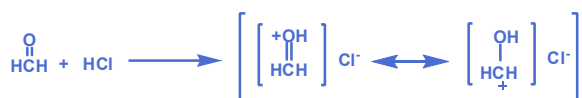


Getteman—Koch反应

38



亲电试剂的形成:

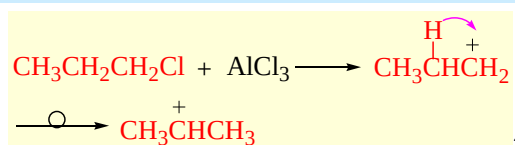
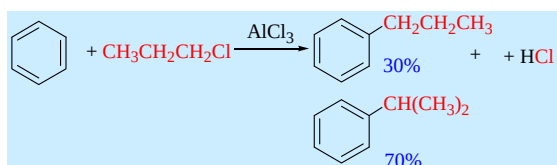


亲电试剂的形成:



Friedel—Crafts反应局限性

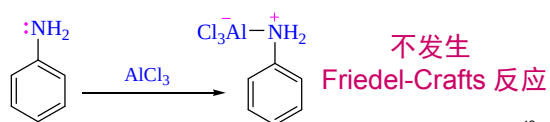
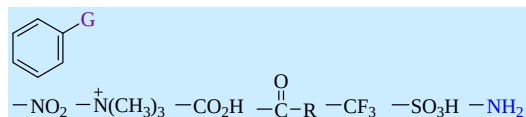
- 当RX超过3个碳原子时，可能发生重排反应



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Friedel—Crafts反应局限性

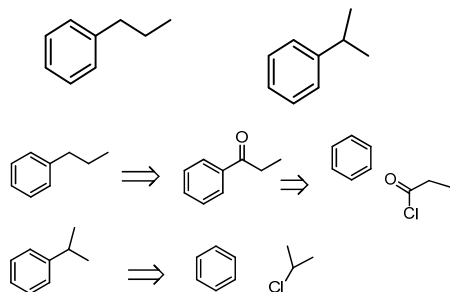
- 当苯环上有强吸电子取代基时，不发生Friedel—Crafts反应



42

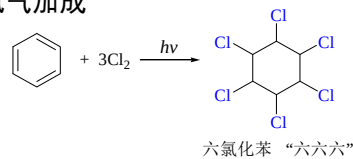
测试

合成下面化合物

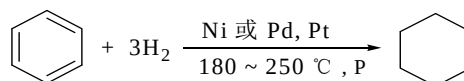


加成与还原反应

- 与氯气加成



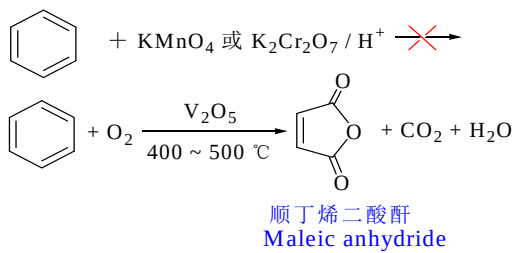
- 加氢



44

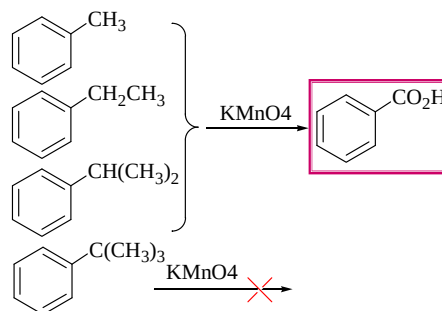
氧化反应

• 苯环的氧化

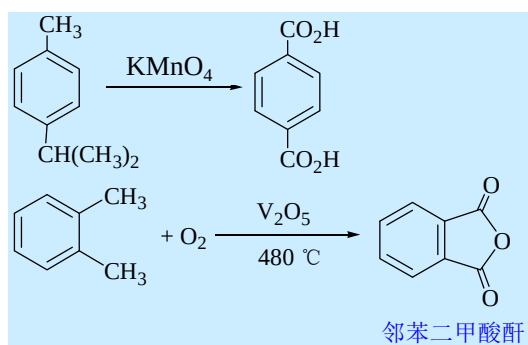


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• 侧链氧化

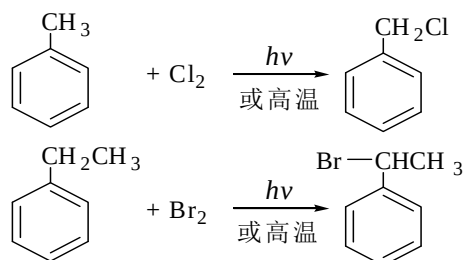


46



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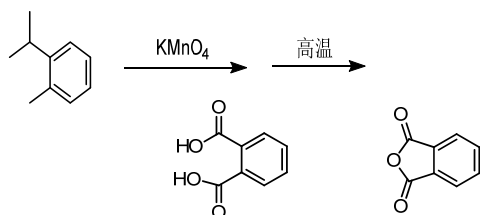
α -卤代反应



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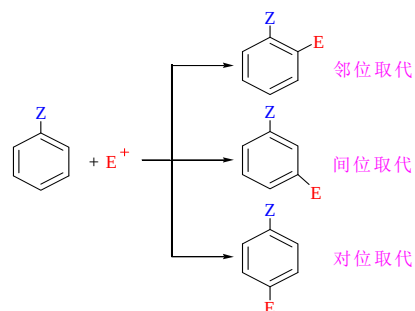
测试

写出下列产物



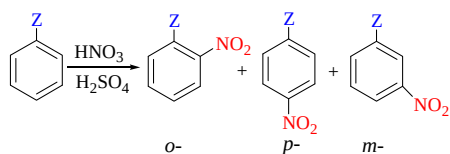
6.4 苯环上亲电取代的定位规律

6.4.1 定位效应(Orientation)



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定位效应

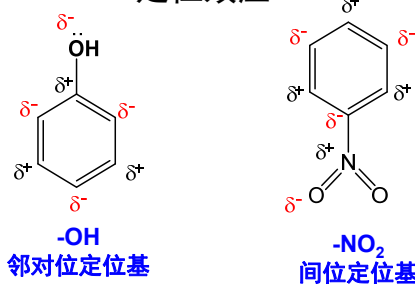


Z=	OH	Me	Cl	NO_2	CN
o(%)	40	58	30	6	17
p(%)	60	38	69	<1	2
m(%)	<1	4	1	93	81

定位效应: 取代基进入的位置受原有取代基的影响

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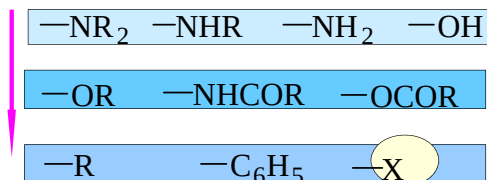
定位效应



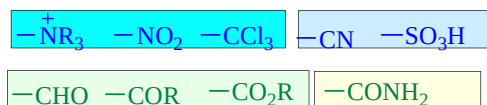
本质: 外来亲电试剂选择性进攻碳 δ^-

52

第一类定位基(邻对位定位基) **给电子取代基 (除X外)**

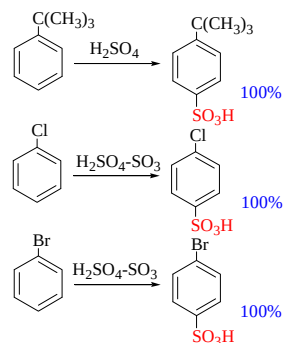


第二类定位基(间位定位基) **吸电子取代基**



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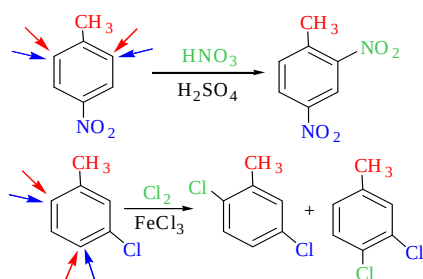
取代基与试剂体积都大时，邻位产物极少



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二元取代苯的定位

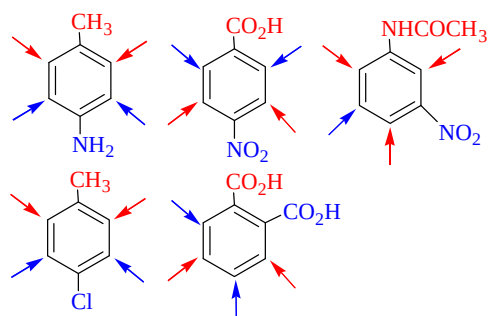
1. 两个取代基定位效应一致



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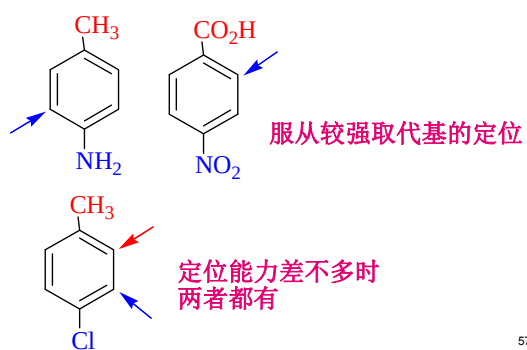
二元取代苯的定位

2. 两个取代基定位效应不一致



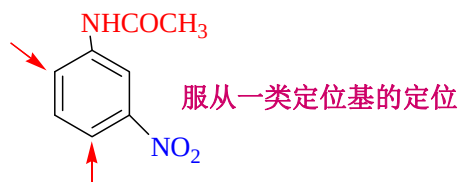
56

两个取代基同类



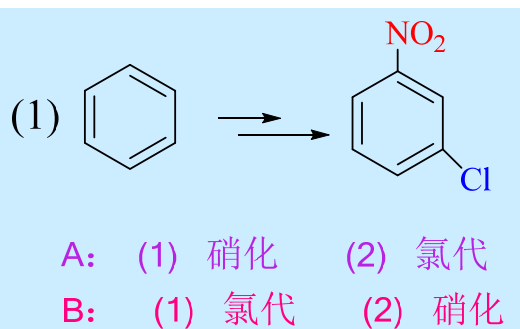
57

两个取代基不同类

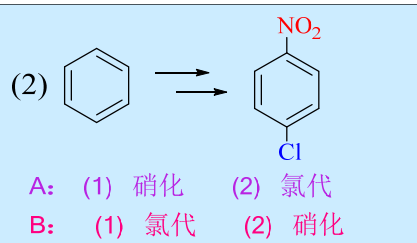
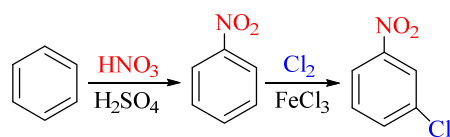


58

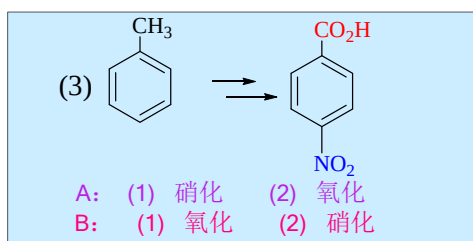
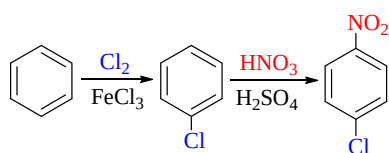
定位规律的应用



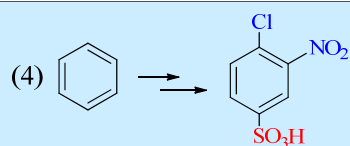
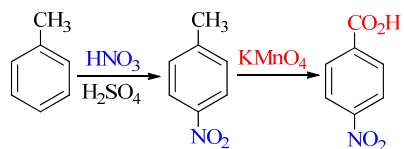
59



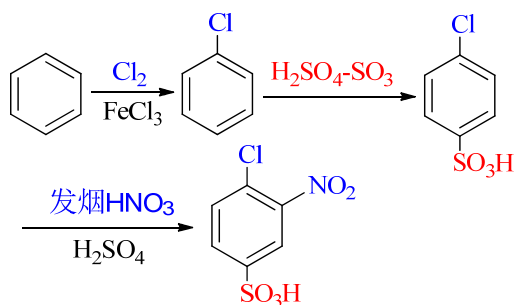
60



61



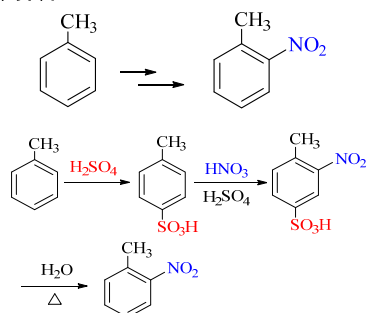
A: (1) 硝化 (2) 氯代 (3) 磺化
 B: (1) 氯代 (2) 硝化 (3) 磺化
 C: (1) 硝化 (2) 磺化 (3) 氯代
 D: (1) 氯代 (2) 磺化 (3) 硝化



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测试

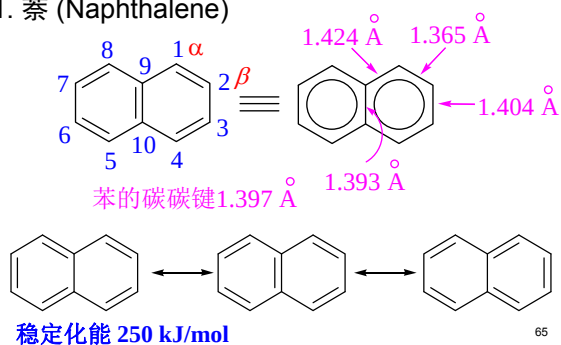
完成下列转化



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稠环芳烃

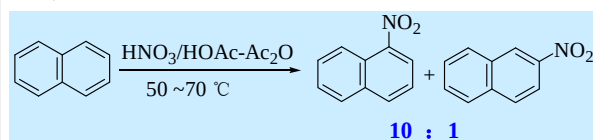
1. 萘 (Naphthalene)



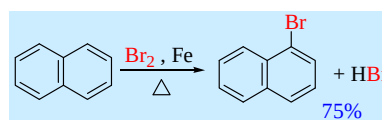
65

2. 萘的反应—亲电取代反应

• 硝化

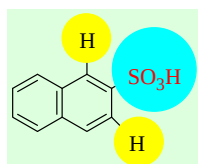
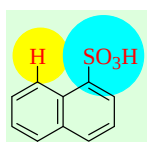
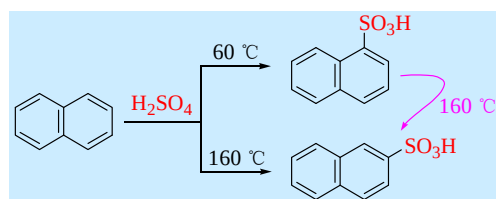


• 卤代



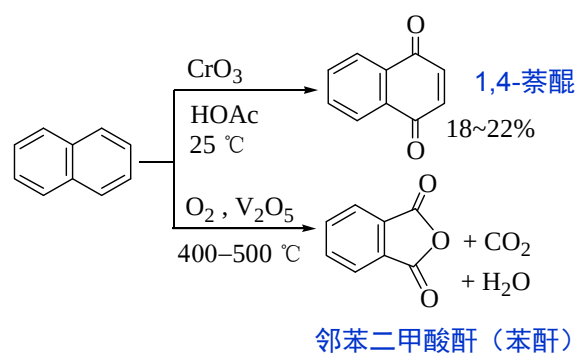
66

• 磺化

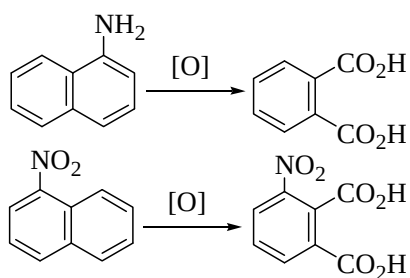


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3. 萘的反应——氧化反应



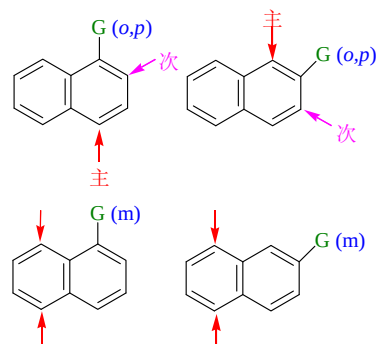
4. 取代萘的氧化反应



电子云密度较高的环被氧化

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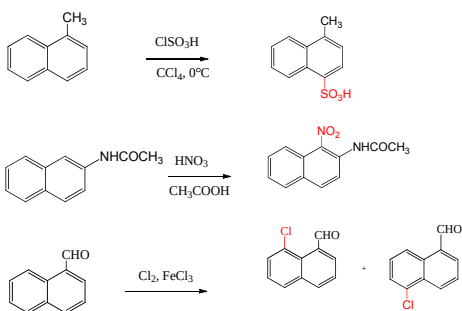
5. 取代萘亲电取代反应的定位



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完成下列转化

测试



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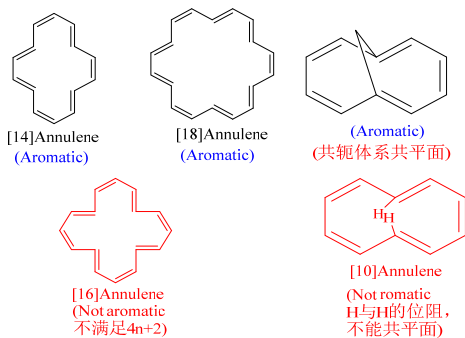
芳香性规则

Hückel's Rule

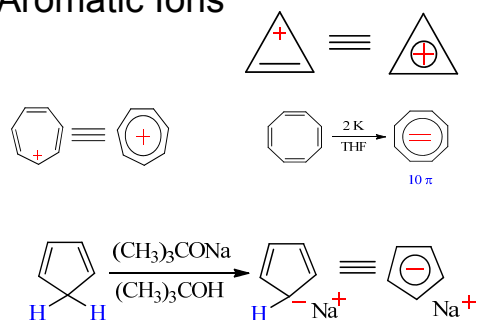
- 共平面
- 共轭体系
- $(4n + 2) \pi$ electrons, n 为整数
- 在化学性质上具有特殊的稳定性, 易于起取代反应, 难于起加成反应

这些特性称为芳香性.

The Annulenes (轮烯)

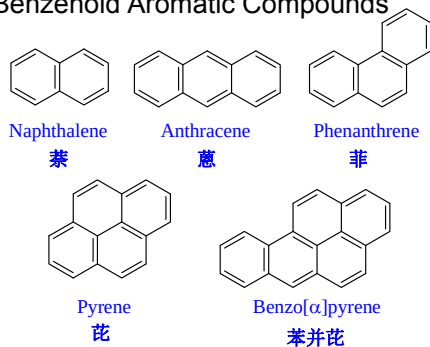


Aromatic Ions



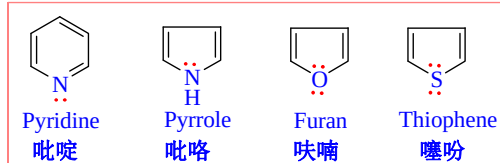
Other Aromatic Compounds

Benzenoid Aromatic Compounds



Heterocyclic Aromatic Compounds:

- Heterocyclic compounds containing N, O or S are by far the most common.



课本练习

P90: 6.3

P90: 6.4

P93: 6.5

P97: 6.9

P104: 2, 3, 4, 5, 7.

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作业 (一)

1. 写出如下反应机理

(1)

Reaction (1) shows toluene (a benzene ring with a methyl group, CH_3) reacting with HNO_3 and H_2SO_4 to produce p-nitrotoluene (a benzene ring with a methyl group and a nitro group, NO_2 , in the para position) and H_2O .

(2)

Reaction (2) shows benzene (a simple benzene ring) reacting with cyclopentyl bromide (a cyclopentane ring attached to a CH_2Br group) and FeBr_3 to form cyclopentylbenzene (a benzene ring attached to a cyclopentane ring).

2. 推断共振效应的电子分布

Reaction 2 shows two molecules: anisole (a benzene ring with a methoxy group, OCH_3) and benzenesulfonic acid (a benzene ring with a sulfonic acid group, SO_3H).

3. 完成下列反应

A

Reaction A shows styrene (vinylbenzene) reacting with H^+ and Fe to form a cyclohexadienyl cation intermediate (a benzene ring with a $\text{CH}_2\text{CH}_2\text{OH}$ group).

B

Reaction B shows the cyclohexadienyl cation intermediate reacting with SO_3 and H_2SO_4 to form a sulfonated cyclohexadienyl cation intermediate (a benzene ring with a $\text{CH}_2\text{CH}_2\text{OH}$ group and a SO_3H group).

C

Reaction C shows the cyclohexadienyl cation intermediate reacting with SO_3 and H_2SO_4 to form a sulfonated cyclohexadienyl cation intermediate (a benzene ring with a $\text{CH}_2\text{CH}_2\text{OH}$ group and a SO_3H group).

D

Reaction D shows the cyclohexadienyl cation intermediate reacting with SO_3 and H_2SO_4 to form a sulfonated cyclohexadienyl cation intermediate (a benzene ring with a $\text{CH}_2\text{CH}_2\text{OH}$ group and a SO_3H group).

E

Reaction E shows the cyclohexadienyl cation intermediate reacting with SO_3 and H_2SO_4 to form a sulfonated cyclohexadienyl cation intermediate (a benzene ring with a $\text{CH}_2\text{CH}_2\text{OH}$ group and a SO_3H group).

作业 (二)

1. 下列那些物质有芳香性

2. 完成下列反应

A $\xrightarrow{\text{AlCl}_3}$

B $\xrightarrow{\text{AlCl}_3}$

C $\xrightarrow{\text{Br}_2, h\nu}$ C

D $\xrightarrow{\text{Br}_2, \text{Fe}}$ D

E $\xrightarrow{\text{KMnO}_4}$ E

F $\xrightarrow[\text{heat}]{\text{KMnO}_4}$

3. 用箭头表示发生亲电取代反应的位点

4. 用苯合成下列物质

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